

Take Home Exercise 3

ECF5410 - Applied Microeconomics

Before answering the questions, take a second to make sure you have all required libraries installed. Run the following lines of code to install the required libraries *if* not already installed.

```
if(!require(tidyverse)) install.packages("tidyverse")
if(!require(modelsummary)) install.packages("modelsummary")
if(!require(fixest)) install.packages("fixest")

library(tidyverse)
library(modelsummary)
library(fixest)
```

For this exercise, you will use the `college-town-rents.csv` in the `data` folder. The data covers rental prices, student population and other variables for 64 US college towns for the years 1980 and 1990.

A table with variable explanations is provided below:

Variable	Description
city	City indicator, 1 to 64
year	80 or 90
pop	City population
enroll	Number of enrolled college students
rent	Average rent
rnthsg	Renter occupied units
tothsg	Occupied housing units
avginc	Average income per capita

Assume that the *population* regression model can be written as:

$$\log(\text{rent}) = \beta_0 + \beta_1 \cdot \log(\text{pctstu}) + \beta_2 \cdot \log(\text{avginc}) + \alpha_i + \epsilon_{it}$$

where *pctstu* is the student population as a percentage of city population (Student Population/City Population $\times 100$) and α_i is the unobserved (time-invariant) city-specific effect. We are interest in the how a stronger presence of students in a city affects the rental prices.

Question 1

Q1.1: Load in the dataset `college-town-rents.csv` and generate the variable “pctstu” (student population as a percentage of city population).

Q1.2: Create the (natural) logarithm for each variable in the population model and report the average value for each of the three variables.

Hint: If you want to apply the same transformation to multiple variables, you *can* use `across()`.

Potential Answer for Question 1:

```
rental <- read_csv("data/college-town-rents.csv") %>%
  mutate(
    pctstu = enroll/pop*100,
    across(
      c(rent, pctstu, avginc),
      ~log(.x),
      .names = "log_{.col}" # create new variables each starting with the prefix "log_"
    )
  )

rental %>%
  summarise(
    across(
      starts_with("log_"),
      ~mean(.x)
    )
  )
```

```
## # A tibble: 1 x 3
##   log_rent log_pctstu log_avginc
##   <dbl>     <dbl>     <dbl>
## 1     5.75       3.19       9.79
```

Question 2

Q2.1: Estimate the equation using pooled OLS. What is your estimate for $\hat{\beta}_1^{\text{POLS}}$?

Potential Answer:

```
pols <- fixest::feols(log_rent ~ log_pctstu + log_avginc, data = rental)
etable(pols)
```

```
##                               pols
## Dependent Var.:              log_rent
##
## Constant          -3.306*** (0.4293)
## log_pctstu         0.1482*** (0.0291)
## log_avginc         0.8765*** (0.0412)
## -----
## S.E. type              IID
## Observations              128
## R2                      0.78425
## Adj. R2                 0.78080
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\hat{\beta}_1^{\text{POLS}} = 0.1482$$

Q2.2: Name *at least one* time-invariant city-specific unobserved confounder and clearly state how you expect this unobserved confounder to be linked with the (1) student population and (2) rental prices?

Potential Answer: One such confounder could be the public transportation network (i.e. subway, trains etc.), which is unlikely to change much over a decade. Better public transport makes a city more attractive

to students who often cannot afford a car. At the same time, cutting down commute time makes it more attractive to move to the suburbs and out of the city which relieves the pressure on rents. Another factor would be political culture: a liberal, non-discriminatory political culture makes the city more attractive to students but also to other potential wealthy renters, driving the price up.

Question 3

Q.3.1: Now estimate the *fixed effects* regression model. What is your estimate for $\hat{\beta}_1^{\text{FE}}$ now?

Potential Answer:

```
fe <- fixest::feols(log_rent ~ log_pctstu + log_avginc | city, data = rental)
etable(fe)
```

```
##                                fe
## Dependent Var.:              log_rent
##
## log_pctstu          0.5049** (0.1678)
## log_avginc          0.9840*** (0.0220)
## Fixed-Effects: -----
## city                                Yes
## -----
## S.E.: Clustered              by: city
## Observations                128
## R2                          0.94999
## Within R2                   0.93228
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\hat{\beta}_1^{\text{FE}} = 0.5049$$

Q.3.2: Does the relative size of the student population appear to affect rental prices?

Potential Answer: Yes, the coefficient is still significant and positive. However, there might be unobserved time-variant variables that confound the estimated relationship, preventing a causal interpretation.

Question 4

Hints:

- First-differences are easiest to calculate if the data is in wide format (also think about which variables in the data you need and drop the remainder to make things easier during pivoting)
- Think about if you need to include an intercept or not in the first-difference regression.
- If you knit your .Rmd file to pdf, use the option `output = "kableExtra"` to safeguard against potential errors. Alternatively, try to knit to `word_document` instead of `pdf_document`.

Q4.1: Now, difference the equation (first-difference) and estimate by OLS. Compare your estimate of $\hat{\beta}_1^{\text{FD}}$ from the fixed effects model in Question 3.

Potential Answer:

```
rental_wide <- rental %>%
  # only select variables we
  select(city, year, starts_with("log_")) %>%
```

```

pivot_wider(
  names_from = "year",
  values_from = starts_with("log_")
) %>%
mutate(
  fd_log_rent = log_rent_90 - log_rent_80,
  fd_log_pctstu = log_pctstu_90 - log_pctstu_80,
  fd_log_avginc = log_avginc_90 - log_avginc_80
)

fd <- fixest::feols(fd_log_rent ~ 0 + fd_log_pctstu + fd_log_avginc, data = rental_wide)

```

Q4.2: Recreate the table above using `modelsummary()`.

Potential Answer:

```

var_labels <- c(
  "fd_log_pctstu" = "Log(Student population share)",
  "fd_log_avginc" = "Log(Average income per capita)",
  "log_pctstu" = "Log(Student population share)",
  "log_avginc" = "Log(Average income per capita)"
)

modelsummary::modelsummary(
  list(
    "POLS" = pols,
    "FE" = fe,
    "FD" = fd
  ),
  coef_map = var_labels,
  gof_map = c("nobs", "r.squared", "adj.r.squared"),
  output = "kableExtra"
) %>%
kable_styling(latex_options = "HOLD_position")

```

	POLS	FE	FD
Log(Student population share)	0.148 (0.029)	0.505 (0.168)	0.505 (0.165)
Log(Average income per capita)	0.876 (0.041)	0.984 (0.022)	0.984 (0.038)
Num.Obs.	128	128	64
R2	0.784	0.950	-0.987
R2 Adj.	0.781	0.898	-1.019

Q4.3: Do you notice any differences between fixed effects and first-difference estimates?

Potential Answer: The standard errors for the fixed effects and first-difference estimator are different because `feols()` does not know that the variables in the first-difference estimation used already additional degrees of freedom by differencing. While `feols()` automatically adjusts the degrees of freedom for the fixed effects estimator here, it does not so for the first-difference estimator.